

Undergraduate Medicine — Physiology Quick Notes

CharaNas — Undergraduate Medicine Department

Study aid only. Confirm with your faculty curriculum and recommended textbooks.

1. Cardiac output = $HR \times \text{Stroke volume}$; $MAP \approx CO \times SVR$.
2. Frank–Starling: $\uparrow$ venous return $\rightarrow \uparrow$ EDV $\rightarrow \uparrow$ stroke volume (within limits).
3. ADH acts on V2 receptors in collecting duct $\rightarrow$ aquaporin-2 insertion $\rightarrow$ water reabsorption.
4. Oxygen–hemoglobin dissociation: right shift ($\uparrow$ unload) with $\uparrow$ CO₂, $\uparrow$ H⁺, $\uparrow$ 2,3-BPG, $\uparrow$ temp.
5. GFR is estimated clinically by creatinine clearance / eGFR equations.